

Mapping Numericals

Q. A digital computer has memory with $64K \times 16$ bit and a cache memory of 1K words. The cache has direct mapping with a block size of 4 words.

- with a block size of 4 words.
- How many bits are there in tag, index & word fields address formula.
 - How many bits are there in the cache & how are they divided into functions & include a valid bit
 - How many blocks can be accommodated
- as $64K \times 16$

89 Main memory can be represented as $64K \times 16$
MM can store $64K$ words of 16 bits each

So main memory size = 64K words

$$= 2^6 \times 2^{10} = 2^{16} \text{ words}$$

Address = 16 bits

Main memory word size = 16 bits each

Data = 16 bits

Cache Memory = 1K Words

= 1024 words

$= 2^{10}$ words

Index = 10 bits

Tag = 16 - 10 = 6 bits

Gracey 1 Block = 4 words.

Total words in cache memory = 1024 words

No of blocks in cache memory = $\frac{1024}{4}$
= 256 blocks
= 2⁸ bits

Block = 8 bits

